# Solution Architecture

HDI Prediction System — Project Design Phase

## Architecture Overview

The system follows a simple three-tier architecture: a data/model layer for training and storing the prediction model, a Flask application layer that serves predictions, and a web-based presentation layer for user interaction. The complete application is deployed on Render.com.

## System Components

| Layer | Component | Description |
|---|---|---|
| Data Layer | hdi.csv Dataset | 2021 global HDI indicators for ~195 countries |
| Model Layer | Linear Regression Model | Trained in HumDevIndex.ipynb; $R^2$ approx. 0.98 |
| Model Layer | Serialized Model (.pkl) | Trained model persisted for inference |
| Application Layer | Flask App | Handles form input, loads model, returns prediction |
| Presentation Layer | HTML/CSS Templates | Dark-themed responsive UI for input and results |
| Deployment Layer | Render.com | Hosts the Flask application for public access |

## Data Flow

- User submits four indicator values through the web form.
- Flask receives the request and validates/parses the input fields.
- The serialized Linear Regression model loads and generates a predicted HDI value.
- Flask renders the result back to the user through the same dark-themed template.

## Model Details

- Algorithm: Linear Regression.
- Features: Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita.
- Target: Human Development Index (HDI).
- Training data: 2021 global indicators for approximately 195 countries.
- Performance: $R^2$ score of approximately 0.98 on the trained data.

## Deployment Architecture

The Flask application, along with its templates and serialized model file, is pushed to a GitHub repository and connected to Render.com for continuous deployment. Render builds and hosts the application, exposing it via a public URL that end users can access directly from a browser.

## Technology Stack

- Language: Python.
- ML Library: scikit-learn (Linear Regression).
- Backend Framework: Flask.
- Front-end: HTML, CSS (dark theme).
- Version Control: Git and GitHub.
- Deployment Platform: Render.com.